

$$\text{Torque} = 5.0(2)(0.5 \sin 40^\circ) = 3.2 \text{ N m}$$

- 7 A** Consider the vertical component of the tension, and letting the angle that the cord makes with the metal sheet be θ ,

$$2T \sin(\theta) = 10 \rightarrow \sin(\theta) = 0.25 \rightarrow \theta = 14.5^\circ$$

$$\text{Length of cord required} = 2(10/\cos(\theta)) = 20.7 = 21 \text{ cm}$$

- 8 C** Option A and D have the wrong units and can be ruled out immediately. Option B is wrong as it